

THE LARGE INTESTINE (Surgery)

Anatomy

- Caecum → rectum (~1.5 m). Features: **teniae coli**, **haustra**, **appendices epiploicae**. **Caecum, transverse, sigmoid = intraperitoneal; ascending + descending = retroperitoneal.**
- **Arterial: SMA → ileocolic, right, middle colic; IMA → left colic, sigmoid, superior rectal.** Venous → portal. **Watershed areas: Griffith's (splenic flexure) + Sudeck's points.**
- Lymphatics: epiploic → paracolic → intermediate → main → para-aortic.

Investigations

- Labs: CBC, CRP, LFTs, **CEA**. **Colonoscopy + biopsy = gold standard.** CT abdomen/pelvis (diverticulitis, staging), contrast enema (**apple-core lesion**), MRI pelvis (rectal cancer), PET-CT (mets).

Diverticular disease

- **Diverticulosis:** mucosal outpouchings, **usually sigmoid**. Risk: age, **low-fibre diet, constipation**. Usually asymptomatic.
- **Acute diverticulitis:** inflammation/microperforation. **LLQ pain, fever, leukocytosis.** Complications: abscess, fistula, perforation, peritonitis, stricture.
- **Management:** uncomplicated → bowel rest, IV fluids, **antibiotics (ceftriaxone + metronidazole)**. Complicated: abscess → drainage; **perforation → Hartmann's**; fistula/stricture → elective sigmoid colectomy.

Colorectal carcinoma

- **95% adenocarcinoma.** Risk: age, adenomas, family history, diet, IBD. Common sites: **rectosigmoid, ascending colon. Adenoma-carcinoma sequence.**
- Types: adenocarcinoma (commonest), carcinoid, lymphoma, GIST, benign adenomas.
- **Clinical: right-sided = occult bleeding + anaemia; left-sided = change in habit + obstruction; rectal = bleeding, tenesmus, incomplete evacuation.**
- **Diagnosis: colonoscopy + biopsy (definitive);** CT chest/abdomen/pelvis (staging); MRI rectum (local); CEA (baseline/follow-up).
- **Surgical management: oncologic resection + lymphadenectomy.** Procedures: **right/left hemicolectomy** (right = ileocolic + right colic vessels; left = left colic ligation), **sigmoid colectomy, anterior resection / APR.** Laparoscopic preferred. **Adjuvant chemo for stage III.** Temporary loop ileostomy/colostomy for diversion.
- **Prognosis: stage I >90% 5-yr; stage III ~50%.** Surveillance: colonoscopy, CEA, CT every 6–12 months.

High-yield one-liners

- **SMA (ileocolic/right/middle colic), IMA (left colic/sigmoid/superior rectal); watershed = Griffith's + Sudeck's.**
- **Diverticulosis = sigmoid, low-fibre; diverticulitis = LLQ pain + fever; perforation → Hartmann's.**
- **CRC = 95% adenocarcinoma; right = anaemia, left = obstruction; colonoscopy + biopsy definitive.**
- **Right/left hemicolectomy by site; adjuvant chemo for stage III; stage I >90% survival.**

THE LARGE INTESTINE (Surgery) — Revision Table

Bold = high-yield. Dr Karzan Seerwan.

ANATOMY & INVESTIGATION

Topic	Key points
Anatomy	Caecum→rectum; teniae coli, haustra, appendices epiploicae ; transverse/sigmoid intraperitoneal, ascending/descending retroperitoneal
Arterial	SMA → ileocolic, right, middle colic ; IMA → left colic, sigmoid, superior rectal
Watershed	Griffith's (splenic flexure) + Sudeck's points
Investigation	Colonoscopy + biopsy = gold standard ; CT staging; apple-core lesion ; CEA

DIVERTICULAR DISEASE

Topic	Key points
Diverticulosis	Mucosal outpouchings, sigmoid ; low-fibre, constipation
Diverticulitis	LLQ pain, fever, leukocytosis ; abscess/fistula/perforation/stricture
Management	Uncomplicated → ceftriaxone + metronidazole ; perforation → Hartmann's ; fistula → elective colectomy

COLORECTAL CARCINOMA

Topic	Key points
Overview	95% adenocarcinoma ; adenoma-carcinoma sequence; rectosigmoid + ascending
Clinical	Right = occult bleed + anaemia ; left = obstruction ; rectal = tenesmus
Diagnosis	Colonoscopy + biopsy ; CT staging; MRI rectum; CEA
Surgery	Oncologic resection + lymphadenectomy ; right/left hemicolectomy by site; adjuvant chemo stage III
Prognosis	Stage I >90%, stage III ~50% ; surveillance colonoscopy/CEA/CT